

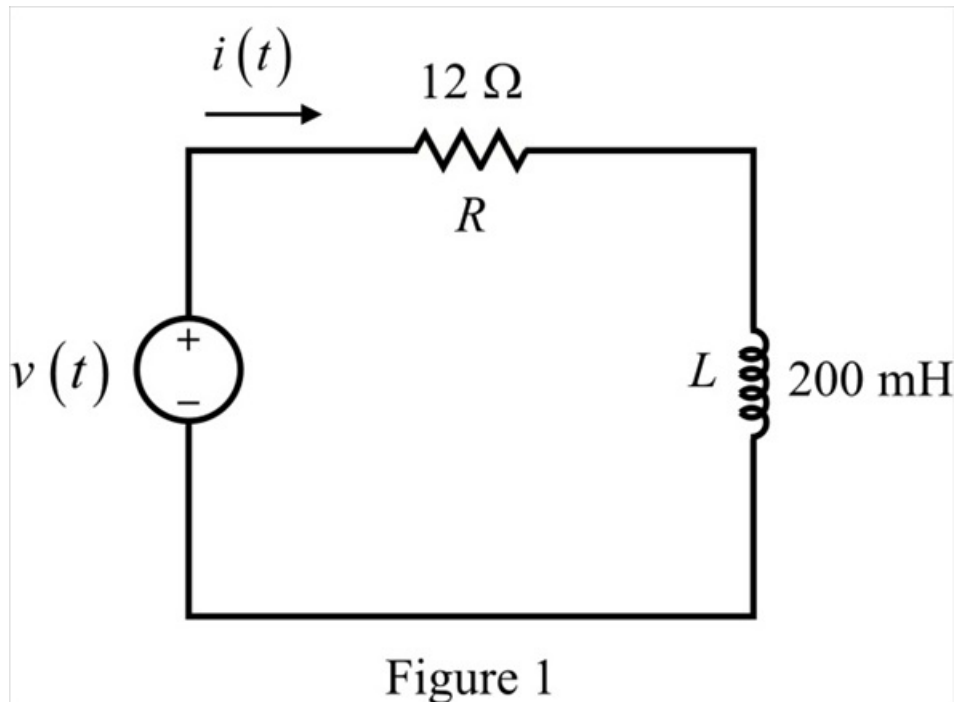
Problem

An electric motor can be modeled as a series combination of a $12\text{-}\Omega$ resistor and 200-mH inductor. If a current $i(t) = 2te^{-10t}$ A flows through the series combination, find the voltage across the combination.

Step-by-step solution

Step 1 of 2

Draw the following circuit:



Step 2 of 2

Apply Kirchhoff's voltage law to the circuit shown in figure 1.

$$v(t) - Ri(t) - L \frac{d}{dt} i(t) = 0$$

Substitute 12 for R , 200×10^{-3} for L .

$$v(t) - 12i(t) - 200 \times 10^{-3} \frac{d}{dt} i(t) = 0$$

$$v(t) = 12i(t) + 0.2 \frac{d}{dt} i(t)$$

Substitute $2te^{-10t}$ for $i(t)$.

$$\begin{aligned} v(t) &= 12[2te^{-10t}] + 0.2 \frac{d}{dt} [2te^{-10t}] \\ &= 24te^{-10t} + 0.2[2e^{-10t} + 2t \times (-10) \times e^{-10t}] \\ &= 24te^{-10t} + 0.2[2e^{-10t} - 20te^{-10t}] \\ &= 24te^{-10t} + 0.4e^{-10t} - 4te^{-10t} \\ &= 20te^{-10t} + 0.4e^{-10t} \text{ V} \end{aligned}$$

Therefore, the voltage across the combination of resistor and inductor is,

$$\boxed{(20t + 0.4)e^{-10t} \text{ V}}.$$